

How to Start the Code:

1. Ensure that you have Java installed on your computer.

<https://www.oracle.com/java/technologies/javase-jdk11-downloads.html>

2. Download and install the Sodium FRP library for Java.

<https://github.com/SodiumFRP/sodium/tree/master/java>

3. Save the provided '**GpsService.java**', '**GpsEvent.java**', '**Main.java**', and '**GpsGUI.java**' files in the same directory.

4. Compile the program with the following command:

```
javac -cp path_to_sodium_jar_file Main.java
```

5. Run the program with the following command:

```
java -cp path_to_sodium_jar_file Main
```

Test Cases

- Test for GPS Data Playback:
 - Run the program and observe the **JTextField** components updating with simulated GPS data.
 - Confirm that the data corresponds to the data in the '**gps.dat**' file.
- Test for Latitude and Longitude Restrictions:
 - Set a latitude and longitude restriction using the **JTextField** components and click the "Set Restriction" button.
 - Confirm that the **JTextArea** component only updates with GPS data within the specified range.
 - Confirm that GPS data outside the specified range is not displayed.
- Test for Total Distance Calculation:
 - Confirm that the total distance traveled by each tracker in the last 5 minutes is displayed and updated accurately.
 - Confirm that the distance calculation only includes data within the specified latitude and longitude range.

These tests are designed to verify that the program is processing and displaying GPS data correctly, adhering to specified restrictions, and accurately calculating the total distance traveled by each tracker.